

SUBHAM MANDAL

+91-9832527356 | subhammandal5555@gmail.com | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

PROFILE

Passionate Electronics & Communication Engineering student and full-stack web developer skilled in building responsive web applications and IoT systems. Strong foundation in software–hardware integration and problem-solving, focused on delivering robust, high-quality solutions.

EDUCATION

Cooch Behar Government Engineering College

Cooch Behar, West Bengal, India

Bachelor of Technology in Electronics & Communication Engineering

2022 – 2026

ACADEMIC ACHIEVEMENTS

- GATE 2026 Qualified – Electronics & Communication Engineering (ECE)

SKILLS

Programming: C, JavaScript

Web Technologies: HTML, CSS, Node.js, Express.js

Embedded & IoT: TI C2000, Code Composer Studio (CCS), Embedded Systems

Tools & Platforms: MATLAB, Git, Netlify, Excel

CERTIFICATIONS

- **Research Intern** at **IIT(BHU)** [View Certificate](#)
- **Full Stack Web Development** by **Delta** [View Certificate](#)
- **Digital Controller for Power Applications** by **IIT MADRAS** [View Certificate](#)

EXPERIENCE

Research Intern at **IIT(BHU)**

June 2025 – July 2025

- Designed and implemented control strategies for different types of converters on the TI C2000 F28379D microcontroller, ensuring efficient power regulation and dynamic response.
- Developed algorithms to coordinate multi-MOSFET switching for a power inverter, achieving high-speed operation with minimized switching losses.
- Integrated PWM generation and closed-loop control techniques for stable and reliable performance in real-time environments.
- Optimized system performance through rigorous testing, debugging, and tuning of power converter hardware and firmware.

CodeSoft

Remote Internship

Web Developer Intern

- Contributed to the development of front-end web projects.
- Tech Stack: HTML, CSS, JavaScript
- GitHub Codebase: [GitHub - Subhammandal5555/CodSoft](#)

PROJECTS

Gyananjali Saraswati Puja Website [Open](#)

Technologies: HTML, CSS, JavaScript, SEO, Payment Integration (UPI), Tawk.to

Description:

- Designed and developed a fully responsive, mobile-first website for Chandra Gyananjali Saraswati Puja (Bankura) with a focus on user experience and accessibility.
- Implemented SEO best practices to improve online visibility and search engine ranking.

- Integrated UPI payment options with clickable buttons and QR code for seamless donations.
- Built interactive features including countdown timer, event schedule, gallery with click-to-open images, and live chat via Tawk.to.
- Optimized site performance and animations for smooth, fast-loading pages.
- Successfully deployed live at gyananjali.online.

Impact / Achievements:

- Enhanced online engagement and facilitated easier donations for the Puja event.
- Provided a centralized platform for attendees to access event details, media, and contact information efficiently.

Personal Portfolio Website [Open](#)

Technologies: Next.js, React, TypeScript, Tailwind CSS, Framer Motion, SVGs, Netlify

Description:

- Designed and developed a high-fidelity, fully responsive personal portfolio website from scratch using Next.js (App Router), TypeScript, and Tailwind CSS.
- Engineered an award-winning style personal website with custom HSL color systems, glow cards, and hand-drawn SVG underline animations.
- Integrated Framer Motion to implement premium dark-mode aesthetics, custom hover effects, glassmorphic panels, and smooth scroll transitions.
- Integrated a secure contact form utilizing Netlify Forms, Google reCAPTCHA, and custom event listeners to auto-dismiss verification overlays on scroll for improved UX.
- Implemented email harvesting protection by obfuscating the contact address in Base64 format to prevent web scraper crawlers from collecting the email for spam.
- Implemented comprehensive SEO optimizations including JSON-LD structured schema markup, robots.ts, sitemaps, and canonical URL mapping for search indexing.
- Successfully deployed live at subhammandal.in.

Impact / Achievements:

- Created a centralized, interactive platform showcasing dual-discipline expertise in full-stack web development and embedded systems engineering.
- Achieved near-perfect (95+) PageSpeed Insights and Core Web Vitals scores by leveraging Next.js font optimization and static asset rendering.

Simon Game

- Created a web version of the Simon game using HTML, CSS, and JavaScript.
- Project Link: <https://simon-subham.netlify.app/>
- Github Link: https://github.com/Subhammandal5555/Simon_Game

Spotify Clone

- Built a frontend clone of the Spotify website using HTML and CSS.
- Project Link: <https://spotify-clone-subham.netlify.app/>
- Github Link: [GitHub - Subhammandal5555/Spotify.clone](https://github.com/Subhammandal5555/Spotify.clone)

ACHIEVEMENTS

- **Community Manager, GDSC CGEC-** grew and managed a 300+ member developer community, organized technical workshops and peer sessions.